Practical exercises — data collection and sampling

A Cambridge insert: what to ask, whom to ask, and how a bad sample ruins a good question.

LESSON 165 | 1 × 40 MIN | MATEMATIKA 7, AMALIY MASHQLAR | S8 6.1–6.2

LESSON CLOCK

8' | 12' | 12' | 6' | 2'

Population and sample	8 min
When a sample misleads	12 min
Writing the question	12 min
Tally and frequency	6 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish a population from a sample.
- Explain why a sample must be representative, and how bias creeps in.
- Write clear survey questions without leading or overlapping options.
- Record data in a tally and frequency table.

TEXTBOOK

National	pp. 499–502
Cambridge	Stage 8, pp. 58–65
Workbook	Exercise 6.1

01 Explanation

Population and sample

TERM	MEANING	EXAMPLE
population	everybody the question is about	all *800* pupils in the school
sample	the part actually asked	*50*
census	asking the whole population	all *800*

WHY SAMPLE AT ALL

A census is exact but slow and expensive. A well-chosen sample of *50* can describe *800* pupils well enough to act on — which is the trade every survey makes.

When a sample misleads

QUESTION	SAMPLE TAKEN	WHAT GOES WRONG
“How much sport does a pupil do?”	the football team	far too much sport
“What is the favourite subject?”	one maths class	maths over-represented
“How do pupils travel to school?”	those in the bus queue	walkers excluded
“Should lessons start later?”	pupils arriving late	the keenest to say yes

A BIG SAMPLE DOES NOT FIX A BIASED ONE

Asking the whole football team instead of half of it makes the wrong answer more confident, not more correct. Bias is repaired by *how* the sample is chosen, not by its size.

Writing the question

FAULT	BAD	BETTER
leading	“Don’t you agree maths is the best subject?”	“Which subject do you like most?”
overlapping options	*0–5, 5–10*	*0–4, 5–9*
gaps in the options	*0–4, 6–9*	*0–4, 5–9*
vague	“Do you read often?”	“How many books did you read last month?”
two questions in one	“Do you like maths and science?”	ask them separately

OPTIONS MUST COVER EVERYTHING, ONCE

Every answer must fit exactly one box: no overlaps, no gaps. That single rule fixes most faulty questionnaires.

Tally and frequency

BOOKS READ LAST MONTH	TALLY	FREQUENCY
0	|||| ||	*7*
1	|||| |||| |||	*13*
2	|||| ||||	*10*
3	|||| |	*6*
4 or more	||||	*4*

The frequencies add to *40* — the size of the sample, which is always worth checking.

TALLY IN FIVES

Four strokes and a cross-stroke make groups that can be counted at a glance. It is a small habit that removes most counting errors from a class survey.

02 Worked examples

EXAMPLE 1 **A school has *800* pupils. A survey asks *50* of them. Name the population and the sample.**

1. The question is about all the pupils.
2. Population: the *800*.
3. Sample: the *50* asked.

ANSWER

Population *800*, sample *50*

EXAMPLE 2 **Why is the football team a poor sample for "how much sport does a pupil do?"**

1. Team members were chosen because they play sport.
2. Their hours are far above average.
3. The sample is biased and overstates the answer.

ANSWER

It is biased

EXAMPLE 3 **What is wrong with the options $0–5$, $5–10$, $10–15$?**

1. A pupil with exactly 5 fits two boxes.
2. The same for 10.
3. Use $0–4$, $5–9$, $10–14$.

ANSWER

The options overlap

03 Interactive model

Run one badly worded question on the class and one careful version of the same question; the difference in the answers makes the lesson better than any list of rules.

Good sample, good question?

Everybody the question is about is the:

- sample
- population
- tally
- frequency

The part actually asked is the:

- sample
- population
- census
- bias

Asking the football team about sport gives:

- a fair answer
- a biased answer
- no answer
- a census

A bigger biased sample is:

- less biased
- just as biased
- unbiased
- a census

“Don’t you agree maths is best?” is:

- clear
- leading
- vague
- fine

Options $0–5$ and $5–10$ are:

- fine
- overlapping
- gapped
- leading

Options $0–4$ and $6–9$ have:

- an overlap
- a gap
- nothing wrong
- too many boxes

Frequencies should add to:

- *1*
- the sample size
- *100*
- the population

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Population	Bosh to‘plam	Генеральная совокупность
Sample	Tanlanma	Выборка
Representative	Vakillik qiluvchi	Репрезентативная
Bias	Siljish	Смещение
Random	Tasodifiy	Случайный
Questionnaire	So‘rovnoma	Анкета
Tally	Sanoq belgisi	Отметка
Frequency	Chastota	Частота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A sample is:

- everybody
- a part of the population
- a question
- a tally

A census asks:

- a sample
- the whole population
- nobody
- ten people

Bias is cured by:

- a bigger sample
- a better chosen sample
- more questions
- rounding

A leading question:

- is clear
- suggests an answer
- has options
- is short

Options must be:

- overlapping
- without gaps or overlaps
- few
- wide

A tally is grouped in:

- twos
- threes
- fives
- tens

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. All *800* pupils are the

   ANSWER population

2. The *50* asked are the

   ANSWER sample

3. Asking everybody is a

   ANSWER census

4. A sample chosen badly gives

   ANSWER bias

5. “Don’t you agree maths is best?” is

   ANSWER a leading question

6. *0–5* and *5–10* are

   ANSWER overlapping

7. Frequencies should add to

   ANSWER the sample size

Medium · the standard question

7 PROBLEMS

1. Why is the football team a poor sample for sport?

   ANSWER They were chosen because they play sport

2. Why is the bus queue a poor sample for travel?

   ANSWER Walkers are excluded

3. Fix the options $0–5$, $5–10$

   ANSWER $0–4$, $5–9$

4. Fix the options $0–4$, $6–9$

   ANSWER $0–4$, $5–9$

5. Rewrite “Do you read often?”

   ANSWER “How many books did you read last month?”

6. Does doubling a biased sample help?

   ANSWER No

7. Frequencies $7, 13, 10, 6, 4$: the sample size

   ANSWER 40

Hard · stretch and reason

7 PROBLEMS

1. Design a fair way to choose *50* from *800* pupils

   ANSWER Number them and draw *50* at random

2. Why not survey only your friends?

   ANSWER They are not representative of the school

3. What is wrong with “Do you like maths and science?”

   ANSWER It asks two questions at once

4. A survey on school lunches asks pupils leaving the canteen: the fault

   ANSWER Those who never eat there are excluded

5. Why must options have no gaps?

   ANSWER Some answers would fit no box

6. A sample of *50* from *800* is what fraction?

   ANSWER $\frac{1}{16}$

7. When is a census worth doing?

   ANSWER When the population is small or the answer must be exact

07 Homework

Homework — 5 tasks

Write your question out fully before asking anybody anything.

1. Write a survey question with five non-overlapping options about time spent reading.
2. Name the population and describe a fair sample for “how do pupils travel to school?”
3. Explain one way a sample can be biased and how to avoid it.
4. Collect *20* answers to your question and record them in a tally table.

5. Find the frequency of each option and check that they add to 20.